Semiconductor Electronics: Materials, Devices and Simple Circuits

Comprehensive Self-Contained Study Tutorial, Theory Synthesis, and Exam Practice

1. Module Context and NEET UG High-Yield Overview

Semiconductor physics constitutes one of the most consistent, highly weighted topics in the NEET UG Physics syllabus. In modern electronic systems, devices that govern a controlled flow of electrons serve as fundamental building blocks. Historically, early radio and electronic equipment utilized **vacuum tubes** (or valves: vacuum diodes, triodes, tetrodes, and pentodes). In these bulky devices, electrons were emitted from a heated cathode via thermionic emission and accelerated through an evacuated space to an anode. Such vacuum devices required high operating voltages (~100 V), consumed significant power for filament heating, had limited lifespans, and broke easily.

Modern solid-state semiconductors revolutionized technology because charge carriers are produced and flow **entirely within the solid crystal lattice itself**. No vacuum or filament heating is necessary. Solid-state devices operate reliably at small DC voltages, feature negligible power consumption, possess long operational lifespans, and allow micro-scale integration.

Key NEET UG Learning Targets:

- Classification of solids based on electrical conductivity (σ) and resistivity (ρ).
- Band theory of solids: Valence Band (VB), Conduction Band (CB), and Energy Band Gap (E_g) across conductors, insulators, and semiconductors.
- Mechanism of electrical conduction in intrinsic semiconductors; the physics of holes and intrinsic carrier concentration (n_i).
- Extrinsic semiconductors: n-type (donor impurities) and p-type (acceptor impurities), energy levels (E_D, E_A), and the Mass Action Law ($n_e n_h = n_i^2$).
- Physics of the p-n junction: diffusion vs. drift currents, depletion layer formation, and barrier potential.
- Semiconductor diode characteristics under forward and reverse bias; dynamic resistance.
- Rectification: Half-wave and full-wave (centre-tapped) rectifiers, output frequencies, and capacitor filter action.

2. Classification of Metals, Conductors, and Semiconductors

A. Classification Based on Conductivity and Resistivity

Solids are broadly grouped according to their room-temperature resistivity (ρ) or electrical conductivity ($\sigma = 1/\rho$):

Category	Resistivity Range (ρ in $\Omega \cdot \mathrm{m}$)	Conductivity Range (σ in $\mathrm{S} \cdot \mathrm{m}^{-1}$)	Characteristic Features
Metals (Conductors)	10^{-2} to 10^{-8}	10^{2} to 10^{8}	Extremely low resistivity; abundant conduction electrons.
Semiconductors	10^{-5} to 10^{6}	10^{5} to 10^{-6}	Intermediate resistivity; sensitive to temperature, light, and dopants.
Insulators	10^{11} to 10^{19}	10^{-11} to 10^{-19}	Extremely high resistivity; tightly bound valence electrons.

BOSONANCE

B. Chemical Classification of Semiconductors

- **Elemental Semiconductors:** Silicon (Si) and Germanium (Ge). Both belong to Group 14 of the periodic table.
- **Compound Semiconductors:**
  - *Inorganic:* Cadmium Sulphide (CdS), Gallium Arsenide (GaAs), Cadmium Selenide (CdSe), Indium Phosphide (InP).
  - *Organic:* Anthracene, doped phthalocyanines.
  - *Organic Conducting Polymers:* Polypyrrole, polyaniline, polythiophene.

C. Classification Based on Energy Bands

In an isolated atom, discrete atomic orbitals define electron energies. When N identical atoms assemble to form a crystal lattice, atomic orbitals overlap. Due to the Pauli exclusion principle, each discrete energy level splits into N closely spaced levels, forming continuous **energy bands**.

- **Valence Band (VB):** The energy band formed by the valence electrons. At absolute zero ($0\,\mathrm{K}$), it is completely filled. The top edge is designated as E_V.
- **Conduction Band (CB):** The lowest unfilled energy band situated directly above the valence band. The bottom edge is designated as E_C. Electrons in this band are mobile and participate in electrical conduction.
- **Energy Band Gap (E_g):** The forbidden energy gap separating the valence band top and the conduction band bottom:

$$E_g = E_C - E_V$$

No permitted electron energy states can exist within this gap.

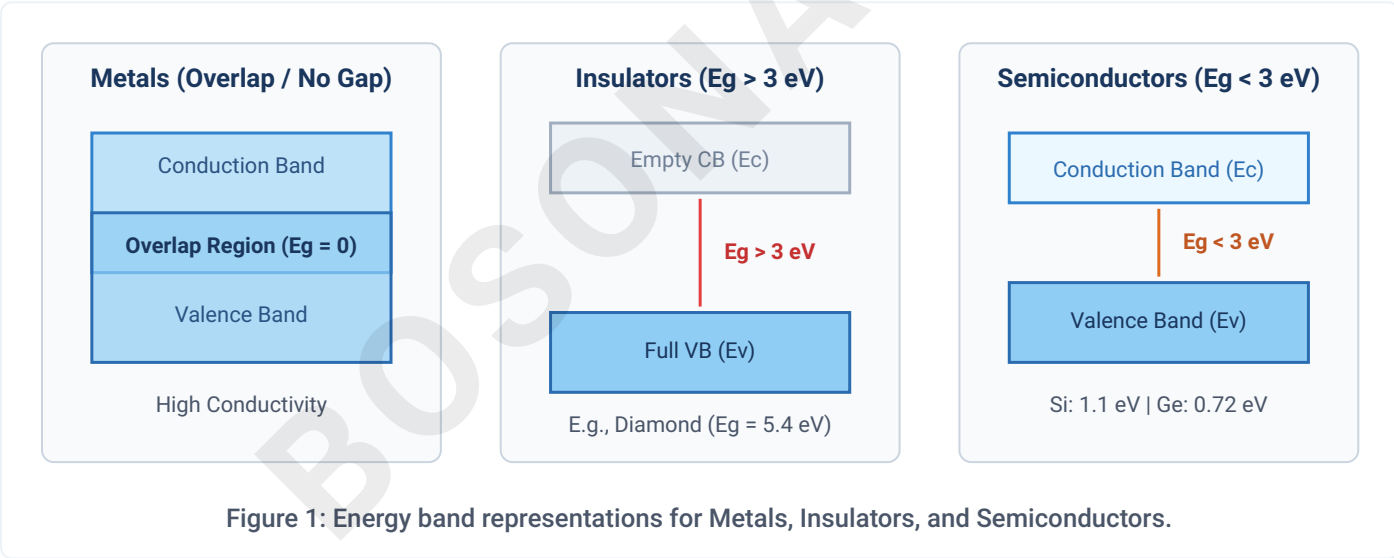


Figure 1: Energy band representations for Metals, Insulators, and Semiconductors.

In crystalline Silicon ($n = 3$ outer shell) and Germanium ($n = 4$ outer shell), each atom has 4 outer electrons. For N atoms, the total number of valence electrons is $4N$, occupying $8N$ outer quantum states ($2s$ and $2p$ subshells). In solid crystal equilibrium, these $8N$ states split into two separate bands of $4N$ states each, separated by E_g:

- The lower band of $4N$ states is fully occupied at $0\,\mathrm{K}$ (Valence Band).
- The upper band of $4N$ states is completely unoccupied at $0\,\mathrm{K}$ (Conduction Band).

Comparative Band Gaps:

$$E_{g,\mathrm{C}}\ (5.4\,\mathrm{eV}) > E_{g,\mathrm{Si}}\ (1.1\,\mathrm{eV}) > E_{g,\mathrm{Ge}}\ (0.7\,\mathrm{eV}) > E_{g,\mathrm{Sn}}\ (0\,\mathrm{eV})$$

Although Carbon (diamond), Silicon, Germanium, and Tin all belong to Group 14, Carbon is an insulator ($E_g > 3\,\mathrm{eV}$), Silicon and Germanium are semiconductors, and Tin is a metal ($E_g = 0\,\mathrm{eV}$).

BOSONANCE

3. Intrinsic Semiconductors

A pure, undoped semiconductor crystal is called an **intrinsic semiconductor**. Silicon and Germanium crystallize in a tetrahedral, diamond-like diamond-cubic lattice where every atom is bonded to four nearest neighbors by shared electron pairs (covalent bonds). Respective lattice constants are $3.56\,\text{Å}$ for Carbon, $5.43\,\text{Å}$ for Silicon, and $5.66\,\text{Å}$ for Germanium.

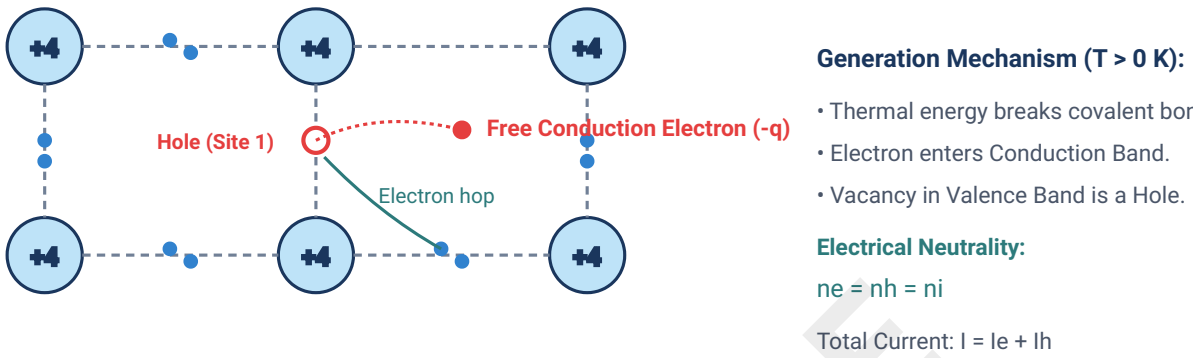


Figure 2: Two-dimensional covalent bonding lattice in Silicon showing thermal generation of an electron-hole pair.

A. Behavior at Absolute Zero ($T = 0\,\text{K}$)

At $0\,\text{K}$, all valence electrons are locked securely inside covalent bonds. Thermal excitation energy is completely absent. Consequently, the valence band is completely full, the conduction band is completely empty, and the crystal behaves as an **ideal electrical insulator**.

B. Thermal Generation and the Nature of a Hole ($T > 0\,\text{K}$)

As temperature rises, lattice vibrations impart thermal energy to valence electrons. When an electron gains energy exceeding E_g, it breaks away from its covalent bond, enters the conduction band, and becomes a mobile **conduction electron** with charge $-q$. It leaves behind a localized vacant bond site with an effective positive charge $+q$, designated as a **hole**.

A hole is not an actual physical subatomic particle; it is an *apparent particle* representing the collective motion of bound valence electrons. When a neighboring valence electron jumps to fill this vacancy, the hole appears to move in the opposite direction. Thus:

- Free conduction electrons move in the conduction band under an electric field, producing electron current I_e.
- Holes move through the valence band towards the negative potential, producing hole current I_h.
- The total current flowing through an intrinsic semiconductor is the direct arithmetic sum:

$$I = I_e + I_h$$

Intrinsic Carrier Balance: In a pure semiconductor, every thermally generated conduction electron creates exactly one hole. Thus:

$$n_e = n_h = n_i$$

where n_e is the conduction electron concentration, n_h is the hole concentration, and n_i is the intrinsic carrier concentration. Dynamic equilibrium ensures that the rate of carrier generation equals the rate of electron-hole recombination.

BOSONANCE

4. Extrinsic Semiconductors

Because the intrinsic carrier concentration of pure Silicon at room temperature is extremely small ($n_i \approx 1.5 \times 10^{16}\,\mathrm{m}^{-3}$ compared to an atomic density of $\sim 5 \times 10^{28}\,\mathrm{atoms \cdot m}^{-3}$), intrinsic semiconductors exhibit very high electrical resistance and cannot be used for practical device fabrication. Their conductivity is enhanced by many orders of magnitude via **doping**.

Doping is the deliberate addition of a precise, minute concentration (typically a few parts per million, ppm) of selected impurity atoms into a pure semiconductor crystal lattice. The added impurities are termed **dopants**. To avoid distorting the parent crystal lattice, dopant atoms must have atomic radii nearly identical to the host Silicon or Germanium atoms.

A. n-Type Semiconductors

When a tetravalent host (Si or Ge) is doped with a **pentavalent impurity** (Group 15 elements: Phosphorus [P], Arsenic [As], or Antimony [Sb]):

- Four of the dopant's five valence electrons establish covalent bonds with four neighboring Silicon atoms.
- The fifth valence electron remains loosely bound to its parent donor nucleus by an extremely small electrostatic ionization energy:

$$\Delta E \approx 0.01\,\mathrm{eV}\ (\text{in Germanium}), \quad \Delta E \approx 0.05\,\mathrm{eV}\ (\text{in Silicon})$$

- At room temperature ($\approx 300\,\mathrm{K}$, where thermal energy $k_B T \approx 0.026\,\mathrm{eV}$), practically all donor atoms are ionised, releasing their fifth electrons into the conduction band.
- The pentavalent atom is termed a **donor impurity** because it donates a conduction electron without generating a hole.
- The donor energy level, E_D, sits just below the bottom of the conduction band (E_C).

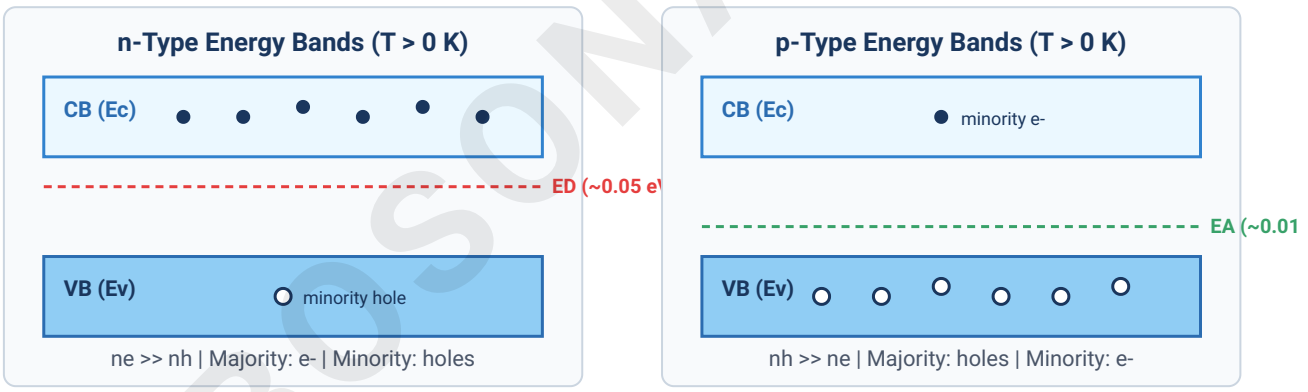


Figure 3: Energy level representations for n-type and p-type semiconductors.

B. p-Type Semiconductors

When Silicon or Germanium is doped with a **trivalent impurity** (Group 13 elements: Boron [B], Aluminium [Al], Indium [In], or Gallium [Ga]):

- The dopant atom provides 3 valence electrons to form bonds with three neighboring host atoms. The fourth bond with the adjacent Silicon atom remains incomplete, creating an inherent hole.
- An electron from a nearby Si—Si covalent bond can easily jump into this vacancy, leaving a hole behind in the host valence band.
- Because the trivalent impurity accepts an electron from the host lattice, it is termed an **acceptor impurity**. Upon accepting an electron, it becomes a fixed, negatively charged immobile ion core.
- The acceptor energy level, E_A, sits slightly above the valence band top (E_V).
- Here, **holes are the majority carriers** and **electrons are the minority carriers** ($n_h \gg n_e$).

BOSONANCE

The Mass Action Law and Overall Crystal Neutrality:

1. Under thermal equilibrium, the product of electron and hole concentrations is invariant for a given semiconductor at a specific temperature, regardless of doping:

$$n_e n_h = n_i^2$$

In an n-type semiconductor with donor concentration N_D, since practically all donors are ionised at room temperature and $N_D \gg n_i$, the electron concentration is $n_e \approx N_D$. From the Mass Action Law, the minority hole concentration is drastically suppressed:

$$n_h = \frac{n_i^2}{N_D}$$

2. **Crucial Exam Truth:** An extrinsic semiconductor (whether n-type or p-type) maintains **complete electrical neutrality**. For an n-type semiconductor, the total negative mobile charge of electrons is perfectly balanced by the total positive charge of immobile ionised donor cores plus minority holes:

$$q(n_e) = q(N_D^+ + n_h)$$

5. The p-n Junction

A p-n junction is created when a single semiconductor single crystal is fabricated such that one region is doped with acceptor impurities (p-type) and the contiguous region is doped with donor impurities (n-type). A p-n junction cannot be formed by mechanically pressing two independent semiconductor pieces together because surface roughness (typically $\sim 100\,\text{Å}$ to $1000\,\text{Å}$) greatly exceeds interatomic spacing (~ 2 to $3\,\text{Å}$), presenting an insurmountable barrier of interfacial discontinuities and oxide traps.

A. Formation Dynamics: Diffusion vs. Drift

1. **Diffusion Process:** Because of the vast concentration gradient across the metallurgical junction, holes diffuse from the p-side to the n-side ($\text{p} \rightarrow \text{n}$), and free electrons diffuse from the n-side to the p-side ($\text{n} \rightarrow \text{p}$). This migration gives rise to a **diffusion current** directed from the p-side to the n-side.
2. **Formation of the Depletion Region:**
   - When an electron diffuses across the junction from n to p, it unmasks an uncompensated, fixed positive donor ion (P^+, As^+) on the n-side.
   - When a hole diffuses from p to n, it leaves an uncompensated, fixed negative acceptor ion (B^-, Al^-) on the p-side.
   - This localized region spanning both sides of the metallurgical interface is stripped bare of mobile charge carriers; it is termed the **depletion region** (or space-charge region). Its physical thickness is on the order of one-tenth of a micrometre ($d \sim 10^{-7}\,\text{m} = 0.1\,\mu\text{m}$).
3. **Internal Electric Field and Drift Current:** The layer of immobile positive ions on the n-side and immobile negative ions on the p-side establishes an internal electric field $\vec{E}_{\text{in}}$ directed strictly from the **n-side to the p-side**. This electric field sweeps thermally generated minority carriers (electrons in the p-side and holes in the n-side) across the junction. This motion constitutes the **drift current** directed from the **n-side to the p-side**.
4. **Thermal Equilibrium:** Initially, the diffusion current is large while the drift current is negligible. As diffusion continues, the space-charge layer widens, augmenting the internal electric field. The drift current increases until it perfectly balances the diffusion current:

$$I_{\text{diffusion}} = I_{\text{drift}} \implies I_{\text{net}} = 0$$

At equilibrium, no net macroscopic current flows across an isolated p-n junction.

BOSONANCE

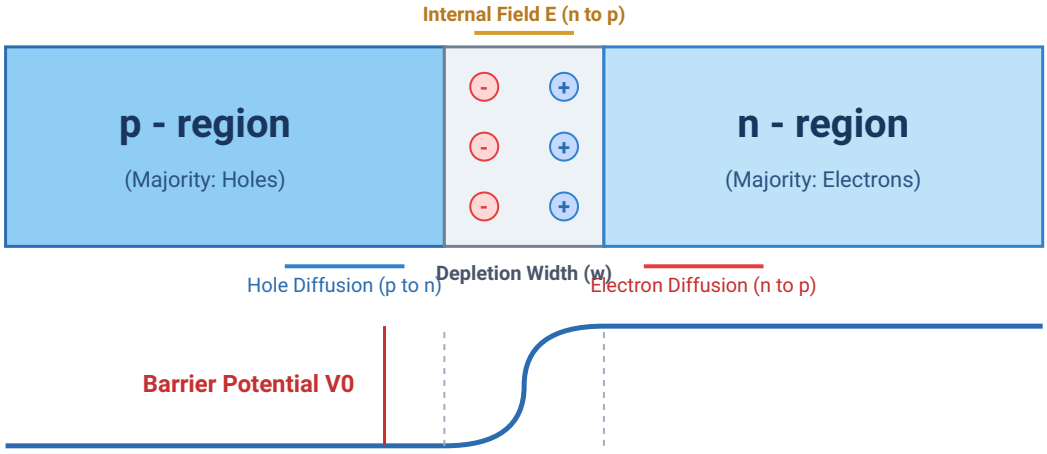


Figure 4: Detailed structure of a p-n junction at equilibrium and the corresponding electrostatic barrier potential profile.

B. Built-In Barrier Potential (V_0)

The separation of fixed charges creates a potential difference across the depletion layer termed the **barrier potential** (V_0). Because the n-region has surrendered electrons while the p-region has accrued electrons, **the n-side resides at a higher electric potential than the p-side**. This potential hill effectively impedes further majority carrier diffusion. At room temperature ($300\,\mathrm{K}$), $V_0 \approx 0.7\,\mathrm{V}$ for Silicon and $V_0 \approx 0.2 \text{ to } 0.3\,\mathrm{V}$ for Germanium.

6. Semiconductor Diode and Biasing

A semiconductor diode is a two-terminal p-n junction device fitted with metallic contacts for connection to an external electrical circuit. In its circuit symbol, the triangle represents the p-type anode (pointing in the direction of conventional forward current flow) and the vertical bar represents the n-type cathode.

A. Forward Bias Condition

A diode is **forward biased** when its p-side is connected to the positive terminal of a battery and its n-side is connected to the negative terminal.

- Because the high-resistance depletion layer bears almost the entire external voltage drop, the externally applied voltage V acts in direct opposition to the built-in barrier potential V_0.
- **Effective Barrier Height:** Decreases from V_0 to $(V_0 - V)$.
- **Depletion Layer Width:** Progressively narrows as majority carriers are repelled by the battery terminals toward the junction.
- **Minority Carrier Injection:** Electrons easily overcome the suppressed barrier, traversing from n to p, while holes traverse from p to n. Upon reaching the opposing sides, they diffuse away from the junction edge as injected minority carriers.
- The forward current is large, measured in milliamperes (mA), and rises exponentially once the applied voltage surpasses the cut-in/threshold voltage ($\sim 0.7\,\mathrm{V}$ for Si, $\sim 0.2\,\mathrm{V}$ for Ge).

B. Reverse Bias Condition

A diode is **reverse biased** when its n-side is connected to the positive terminal of a battery and its p-side is connected to the negative terminal.

- The applied voltage V acts in the same direction as the built-in barrier potential V_0.
- **Effective Barrier Height:** Increases from V_0 to $(V_0 + V)$.
- **Depletion Layer Width:** Widens significantly as mobile holes in the p-side and mobile electrons in the n-side are drawn away from the junction.
- Diffusion of majority carriers is suppressed. The only current that can flow is a minuscule **reverse saturation current** (I_s) generated by the drift of thermally generated minority carriers across the junction.

BOSONANCE

- The reverse current is exceedingly small (microamperes, μA, in Germanium; nanoamperes, nA, in Silicon). Over a wide range of reverse voltage, I_s is practically independent of voltage because it is strictly limited by the rate of thermal generation of minority carriers, not by the applied electric field.
- **Reverse Breakdown Voltage (V_{br}):** At a critical high reverse voltage, covalent bonds rupture or avalanche multiplication ensues, causing a sudden surge in reverse current. Without external current-limiting resistance, the diode will suffer destructive thermal runaway.

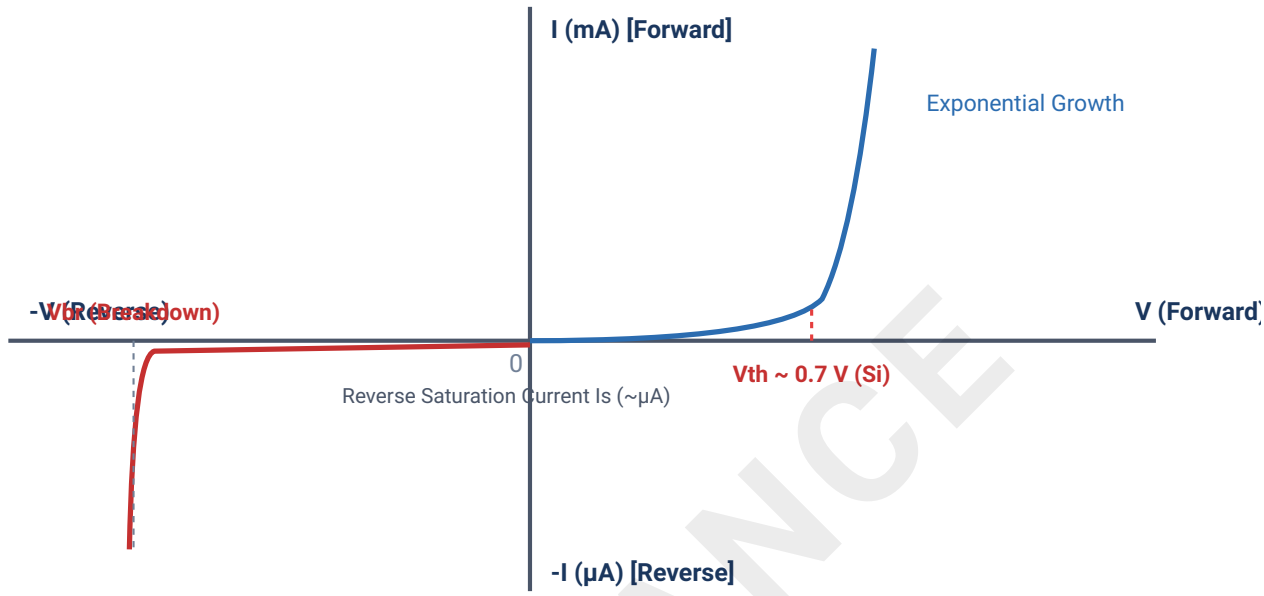


Figure 5: Typical V-I characteristic curve for a Silicon p-n junction diode.

Dynamic (AC) Resistance Formula: The dynamic resistance of a diode at a specific operational operating point is defined as the ratio of a small incremental change in voltage (ΔV) to the resulting incremental change in current (ΔI):

$$r_d = \frac{\Delta V}{\Delta I}$$

In forward conduction beyond the knee voltage, r_d is typically very small ($\sim 1 \text{ to } 30\,\Omega$). Under reverse bias prior to breakdown, r_d is exceptionally large ($\sim 10^6 \text{ to } 10^7\,\Omega$).

7. Applications of Junction Diodes: Rectification

Because a p-n junction diode permits substantial current conduction strictly when forward biased and presents virtually infinite resistance under reverse bias, it acts as an ideal electronic **rectifier**: converting alternating voltages/currents (AC) into unidirectional, pulsating direct voltages/currents (DC).

A. Half-Wave Rectifier

In a half-wave rectifier, an alternating AC voltage from the secondary winding of a transformer is applied across a single diode connected in series with a load resistor R_L.

- **Positive Half-Cycle:** The transformer terminal connected to the diode anode becomes positive with respect to the cathode. The diode is forward biased and conducts. Current flows through R_L, generating an output voltage pulse across it.
- **Negative Half-Cycle:** The transformer terminal connected to the anode becomes negative. The diode is reverse biased and non-conducting. The current through R_L is essentially zero.
- **Output Characteristics:** The rectified output consists of single-polarity half-sinusoids appearing only during alternate half-cycles.

BOSONANCE

- **Fundamental Frequency Relationship:** If the input AC line frequency is f, the ripple frequency of the output pulsating DC is identical:
$$f_{\text{out}} = f_{\text{in}}$$
For an input frequency of $50\ \text{Hz}$, the half-wave rectified output pulses at $50\ \text{Hz}$.

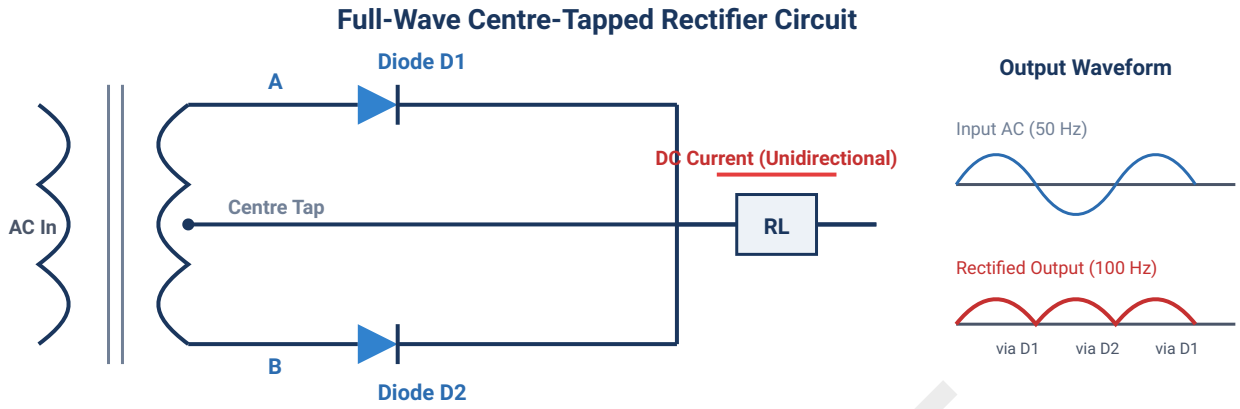


Figure 6: Complete schematic of a full-wave centre-tapped rectifier and input/output waveforms.

B. Full-Wave Rectifier (Centre-Tapped)

To utilize both halves of the input cycle, a centre-tapped transformer with two diodes (D_1 and D_2) is employed:

- The centre tap is treated as the zero-potential reference point.
- **During Positive Half-Cycle:** Terminal A is positive with respect to the centre tap, while Terminal B is negative. Diode D_1 is forward biased and conducts; Diode D_2 is reverse biased and non-conducting. Current flows through the load resistor R_L from terminal X to Y.
- **During Negative Half-Cycle:** Terminal A becomes negative while Terminal B becomes positive with respect to the centre tap. Diode D_1 is reverse biased, while Diode D_2 is forward biased and conducts. Current once again flows through R_L in the *identical direction* from X to Y.
- **Frequency Doubling:** Current flows through the load during both half-cycles. Hence, the output pulsating frequency is exactly double the input AC line frequency:
$$f_{\text{out}} = 2f_{\text{in}}$$
For a $50\ \text{Hz}$ input, the full-wave rectified output ripple frequency is $100\ \text{Hz}$.

C. Capacitor Filter Circuit

The rectified output from half-wave or full-wave circuits consists of pulsating DC with substantial AC ripple. To obtain steady DC suitable for electronic devices, a **smoothing filter** is placed across the output terminals.

- A capacitor of large capacitance C is connected in parallel with the load resistor R_L.
- **Charging Phase:** As the rectified pulse voltage rises toward its peak value V_0, the capacitor charges rapidly (through the low forward resistance of the conducting diode).
- **Discharging Phase:** As the rectifier voltage begins to drop, the diode becomes reverse biased because the capacitor retains a voltage higher than the input transformer voltage. The capacitor discharges through R_L with a characteristic discharge time constant:
$$\tau = R_L C$$
- By choosing a large capacitance C such that $R_L C \gg T$ (where T is the period of the rectified wave), the voltage across R_L decays negligibly before the next rectified voltage pulse charges the capacitor back to peak value.
- Consequently, the output voltage remains smoothed and clamped near the peak AC voltage V_{peak}.

BOSONANCE

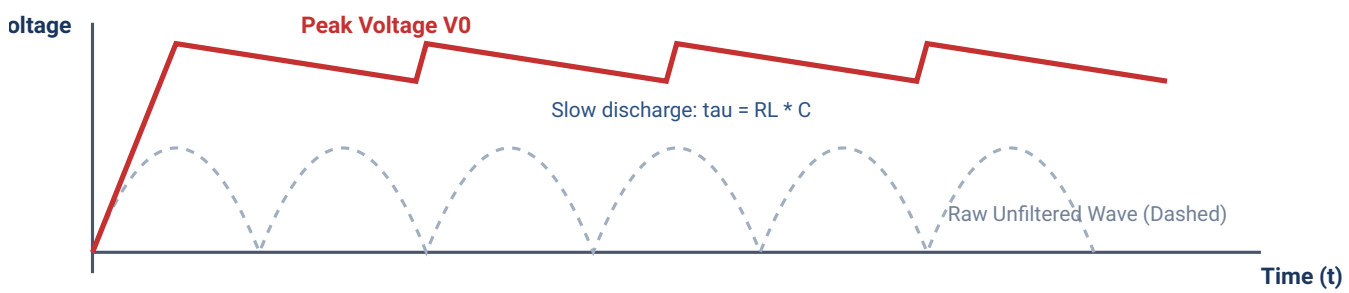


Figure 7: Output voltage waveform of a full-wave rectifier with a parallel capacitor smoothing filter.

8. Rigorous Worked Examples

Worked Example 1: Charge Carrier Concentrations in Extrinsic Silicon

Problem: A pure silicon crystal contains $5 \times 10^{28}\,\text{atoms} \cdot \text{m}^{-3}$. It is doped with a pentavalent Arsenic impurity at a concentration of $1\,\text{ppm}$ ($1\,\text{part per million}$). If the intrinsic carrier concentration of Silicon at room temperature is $n_i = 1.5 \times 10^{16}\,\text{m}^{-3}$, calculate:

1. The majority carrier concentration (n_e).
2. The minority carrier concentration (n_h).

Solution:

Step 1: Calculate donor atom concentration N_D:

$$N_D = \frac{1}{10^6} \times (5 \times 10^{28}\,\text{m}^{-3}) = 5 \times 10^{22}\,\text{atoms} \cdot \text{m}^{-3}$$

Step 2: At room temperature, essentially all donor atoms are ionised. Because $N_D \gg n_i$ ($5 \times 10^{22} \gg 1.5 \times 10^{16}$), the thermally generated electron concentration is completely negligible in comparison to the donor contribution. Hence:

$$n_e \approx N_D = 5 \times 10^{22}\,\text{m}^{-3}$$

Step 3: Using the Mass Action Law under thermal equilibrium ($n_e n_h = n_i^2$):

$$n_h = \frac{n_i^2}{n_e} = \frac{(1.5 \times 10^{16})^2}{5 \times 10^{22}} = \frac{2.25 \times 10^{32}}{5 \times 10^{22}} = 4.5 \times 10^{9}\,\text{m}^{-3}$$

Conclusion: Doping increased the conduction electron concentration from $1.5 \times 10^{16}\,\text{m}^{-3}$ to $5 \times 10^{22}\,\text{m}^{-3}$, while the hole concentration was suppressed through recombination to $4.5 \times 10^{9}\,\text{m}^{-3}$.

BOSONANCE

Worked Example 2: Dynamic Resistance Determination from Experimental Characteristics

Problem: An experimental silicon diode exhibits the following operating points: in forward bias, current changes from $10\,\text{mA}$ at $0.7\,\text{V}$ to $20\,\text{mA}$ at $0.8\,\text{V}$. Under reverse bias, a reverse voltage of $-10\,\text{V}$ results in a reverse current of $-1\,\mu\text{A}$. Determine:

1. The dynamic forward resistance (r_{fb}) in this region.
2. The dynamic reverse resistance (r_{rb}) at $-10\,\text{V}$.

Solution:

Part 1: Forward Dynamic Resistance:

$$\Delta V_F = 0.8\,\text{V} - 0.7\,\text{V} = 0.1\,\text{V}$$
$$\Delta I_F = 20\,\text{mA} - 10\,\text{mA} = 10\,\text{mA} = 10 \times 10^{-3}\,\text{A}$$
$$r_{fb} = \frac{\Delta V_F}{\Delta I_F} = \frac{0.1}{10 \times 10^{-3}} = 10\,\Omega$$

Part 2: Reverse Resistance:

$$r_{rb} = \frac{V_R}{I_R} = \frac{10\,\text{V}}{1\,\mu\text{A}} = \frac{10}{10^{-6}\,\text{A}} = 1.0 \times 10^{7}\,\Omega = 10\,\text{M}\Omega$$

Conclusion: The forward dynamic resistance is extremely low ($10\,\Omega$), while the reverse resistance is exceptionally large ($10\,\text{M}\Omega$), confirming the near-ideal one-way valve behavior of the diode.

9. NEET UG Exam-Style Practice Questions

The following questions are prepared strictly according to the latest NEET UG format and difficulty level, including single-correct MCQs, Assertion-Reason pairs, and statement-based questions.

Practice Question 1 (NEET UG Single-Correct MCQ)

In an unbiased, isolated p-n junction at room temperature, mobile holes diffuse from the p-region to the n-region primarily because:

(A) The free electrons in the n-region attract holes across the interface.

(B) The built-in potential barrier actively pulls holes into the n-region.

(C) The concentration of holes in the p-region is substantially higher than in the n-region.

(D) Immobile donor ions on the n-side exert an electrostatic attractive force on holes.

Correct Answer: (C)

Explanation: Diffusion is an entropic, concentration-driven transport process. In a p-type region, holes are majority carriers (high concentration), while in the n-type region, holes are minority carriers (low concentration). Due to this steep concentration gradient, holes naturally diffuse from high concentration (p-side) to low concentration (n-side). Note that the built-in barrier potential opposes, rather than aids, this diffusion.

BOSONANCE

Practice Question 2 (NEET UG Assertion & Reason)

Directions: Select the correct option from the following choices:

- (A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.
- (B) Both Assertion and Reason are true, but Reason is NOT the correct explanation of Assertion.
- (C) Assertion is true, but Reason is false.
- (D) Assertion is false, but Reason is true.

Assertion: An n-type silicon crystal carries zero net macroscopic electric charge and is perfectly electrically neutral.

Reason: For every extra conduction electron provided by a pentavalent donor impurity atom, an uncompensated, fixed positive donor ion core is created in the crystal lattice.

Correct Answer: (A)

Explanation: A common trap for NEET aspirants is assuming that because electrons are the majority carriers in an n-type semiconductor, the crystal must carry a net negative charge. In reality, the semiconductor is prepared from electrically neutral host silicon atoms and electrically neutral pentavalent dopant atoms. When a donor atom donates its 5th electron to the conduction band, it becomes a positively charged ion (P^{+}). Thus, the total mobile negative charge is exactly counterbalanced by the total positive charge of the immobile donor cores and thermally generated minority holes. Assertion and Reason are both true, and the Reason correctly explains the Assertion.

Practice Question 3 (NEET UG Single-Correct MCQ)

Carbon, Silicon, and Germanium all possess four valence electrons each in their outermost atomic shells. Their forbidden energy band gaps are respectively denoted as $(E_g)_C$, $(E_g)_{Si}$, and $(E_g)_{Ge}$. Which of the following relationships is correct?

(A) $(E_g)_{Si} < (E_g)_{Ge} < (E_g)_C$

(B) $(E_g)_C < (E_g)_{Si} < (E_g)_{Ge}$

(C) $(E_g)_C > (E_g)_{Si} > (E_g)_{Ge}$

(D) $(E_g)_C = (E_g)_{Si} = (E_g)_{Ge}$

Correct Answer: (C)

Explanation: The valence electrons of Carbon, Silicon, and Germanium reside in the second ($n = 2$), third ($n = 3$), and fourth ($n = 4$) atomic orbits, respectively. With increasing principal quantum number and atomic radius, the valence electrons are bound less tightly to the nucleus, requiring less energy to ionize. Consequently, the energy band gap decreases systematically down the group:

$$(E_g)_C\,(5.4\,\mathrm{eV}) > (E_g)_{Si}\,(1.1\,\mathrm{eV}) > (E_g)_{Ge}\,(0.72\,\mathrm{eV})$$

BOSONANCE

Practice Question 4 (NEET UG Numerical / Frequency Problem)

An alternating sinusoidal voltage having a frequency of $60\,\text{Hz}$ is fed into the primary winding of a transformer. Calculate the fundamental ripple frequency of the output voltage across the load resistor when the secondary is connected to:

1. A half-wave rectifier circuit.
2. A centre-tapped full-wave rectifier circuit.

(A) $60\,\text{Hz}$ in half-wave, and $60\,\text{Hz}$ in full-wave.

(B) $30\,\text{Hz}$ in half-wave, and $60\,\text{Hz}$ in full-wave.

(C) $60\,\text{Hz}$ in half-wave, and $120\,\text{Hz}$ in full-wave.

(D) $120\,\text{Hz}$ in half-wave, and $60\,\text{Hz}$ in full-wave.

Correct Answer: (C)

Explanation:

1. In a half-wave rectifier, current flows only during the positive half-cycle; the output waveform repeats once every full input period T. Therefore, the output frequency equals the input frequency:

$$f_{\text{HW}} = f_{\text{in}} = 60\,\text{Hz}$$

2. In a full-wave rectifier, both positive and negative half-cycles are converted into positive pulses, meaning the output waveform repeats every half-period ($T/2$). The ripple frequency is therefore doubled:

$$f_{\text{FW}} = 2f_{\text{in}} = 2 \times 60 = 120\,\text{Hz}$$

Practice Question 5 (NEET UG Statement Analysis)

Consider the following two statements regarding a p-n junction diode:

- **Statement I:** Applying a forward bias voltage lowers the potential barrier height and narrows the width of the depletion layer.
- **Statement II:** Applying a reverse bias voltage suppresses majority carrier diffusion and causes the reverse current to depend linearly on the magnitude of the applied reverse voltage up to breakdown.

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct, but Statement II is incorrect.

(D) Statement I is incorrect, but Statement II is correct.

Correct Answer: (C)

Explanation:

Statement I is correct: The forward applied voltage opposes the built-in barrier, decreasing effective barrier height to $(V_0 - V)$ and narrowing the depletion region.

Statement II is incorrect: The reverse current (reverse saturation current I_s) is almost completely *independent* of the applied reverse voltage (it is voltage-saturated, not linear). It is limited strictly by the thermal generation rate of minority carriers across the junction boundaries, not by the applied voltage.

BOSONANCE

Practice Question 6 (NEET UG Single-Correct MCQ)

A pure semiconductor material is doped with a trivalent impurity at room temperature. Which of the following statements correctly characterizes the resulting extrinsic semiconductor?

(A) Holes are the majority carriers; the acceptor energy level E_A lies slightly above the top of the valence band.

(B) Electrons are the majority carriers; the donor energy level E_D lies slightly below the bottom of the conduction band.

(C) Holes are the minority carriers; the donor energy level E_D lies midway in the forbidden energy gap.

(D) Electrons are the minority carriers; the acceptor energy level E_A lies slightly below the bottom of the conduction band.

Correct Answer: (A)

Explanation: Doping with a trivalent impurity (e.g., Boron, Indium) produces a p-type semiconductor where holes are the majority carriers ($n_h \gg n_e$). The accepted electrons occupy an acceptor energy level E_A located merely $\sim 0.01 \text{ to } 0.05\,\text{eV}$ directly above the valence band edge E_V, facilitating easy excitation of valence electrons into E_A and leaving mobile holes in the valence band.

10. Common Student Pitfalls and Revision Checklist

NEET UG Critical Revision Checklist & Common Pitfalls:

- **Pitfall 1 (Charge of Extrinsic Crystals):** Do not think that an n-type semiconductor is negatively charged or that a p-type semiconductor is positively charged. *Both are strictly electrically neutral.*
- **Pitfall 2 (Direction of Diffusion vs. Drift):** Remember the exact conventions:
  - **Hole diffusion:** from $\text{p} \rightarrow \text{n}$.
  - **Electron diffusion:** from $\text{n} \rightarrow \text{p}$.
  - **Diffusion current (conventional):** directed from $\text{p} \rightarrow \text{n}$.
  - **Internal electric field:** directed from $\text{n} \rightarrow \text{p}$ (due to positive donor ions on the n-side and negative acceptor ions on the p-side).
  - **Drift current (conventional):** directed from $\text{n} \rightarrow \text{p}$.
- **Pitfall 3 (Mass Action Law Application):** The equation $n_e n_h = n_i^2$ holds under *thermal equilibrium* for both intrinsic and extrinsic semiconductors. Note that n_i is an exponential function of temperature; at a fixed temperature, raising n_e by donor doping forces a sharp decline in n_h through recombination.
- **Pitfall 4 (Nature of Reverse Saturation Current):** Reverse saturation current I_s is insensitive to applied voltage, but it is *highly sensitive to temperature*. Because I_s originates from thermally generated electron-hole pairs, it roughly doubles for every $10\,^\circ\text{C}$ rise in temperature.
- **Pitfall 5 (Rectifier Frequency vs. Filter Ripple):**
  - Half-wave output frequency = f_{in}.
  - Full-wave output frequency = $2f_{\text{in}}$.
  - A capacitor filter connects in *parallel* with the load R_L; an inductor filter connects in *series* with R_L.

BOSONANCE

11. Final Comprehensive Recap

1. Energy Bands: Insulators have $E_g > 3\,\text{eV}$ (Diamond $= 5.4\,\text{eV}$); semiconductors have $E_g < 3\,\text{eV}$ ($\text{Si} = 1.1\,\text{eV}$, $\text{Ge} = 0.72\,\text{eV}$); metals have overlapping bands or partially filled conduction bands ($E_g = 0$).

2. Intrinsic Transport: Pure crystal with $n_e = n_h = n_i$. Total conduction current is $I = I_e + I_h$. Perfect insulator at $0\,\text{K}$.

3. Extrinsic Doping: Pentavalent dopants (P, As, Sb) give n-type semiconductors ($n_e \gg n_h$) with donor level E_D just below E_C. Trivalent dopants (B, Al, In) give p-type semiconductors ($n_h \gg n_e$) with acceptor level E_A just above E_V. Mass Action Law: $n_e n_h = n_i^2$.

4. p-n Junction Mechanics: Formed by diffusion of majority carriers and subsequent drift of minority carriers under the induced electric field. Equilibrium corresponds to $I_{\text{diff}} = I_{\text{drift}} \implies I_{\text{net}} = 0$. Depletion layer comprises uncompensated immobile ions ($w \sim 0.1\,\mu\text{m}$).

5. Diode Biasing: Forward bias reduces barrier to $(V_0 - V)$ and produces mA diffusion currents beyond cut-in ($0.7\,\text{V}$ for Si, $0.2\,\text{V}$ for Ge). Reverse bias increases barrier to $(V_0 + V)$, yielding a minute μA minority drift saturation current.

6. Rectifier Formulas: Half-wave produces pulsating DC at $f_{\text{out}} = f_{\text{in}}$; full-wave produces pulsating DC at $f_{\text{out}} = 2f_{\text{in}}$. Smoothing is achieved with a parallel capacitor filter ($\tau = R_L C$).

BOSONANCE